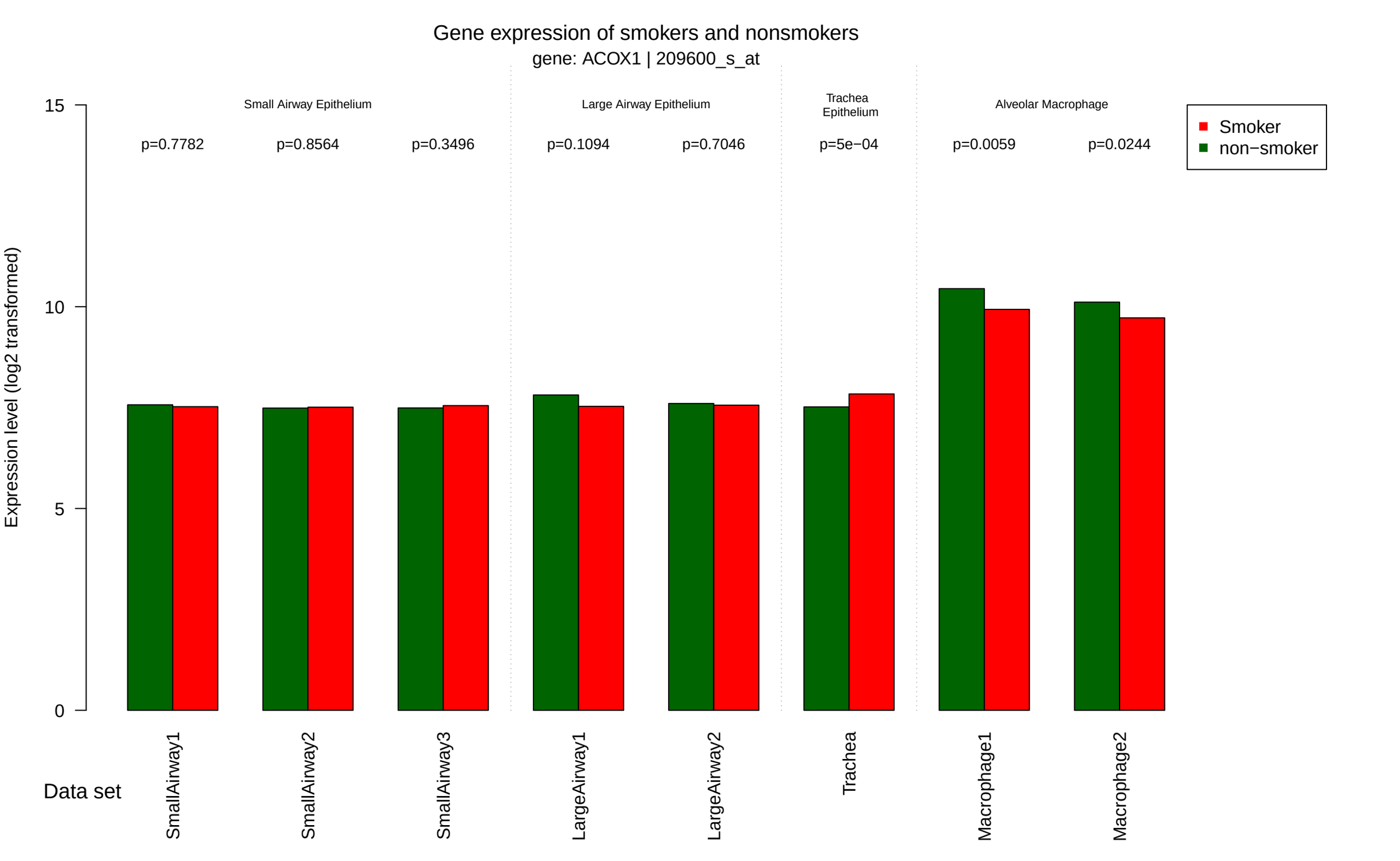
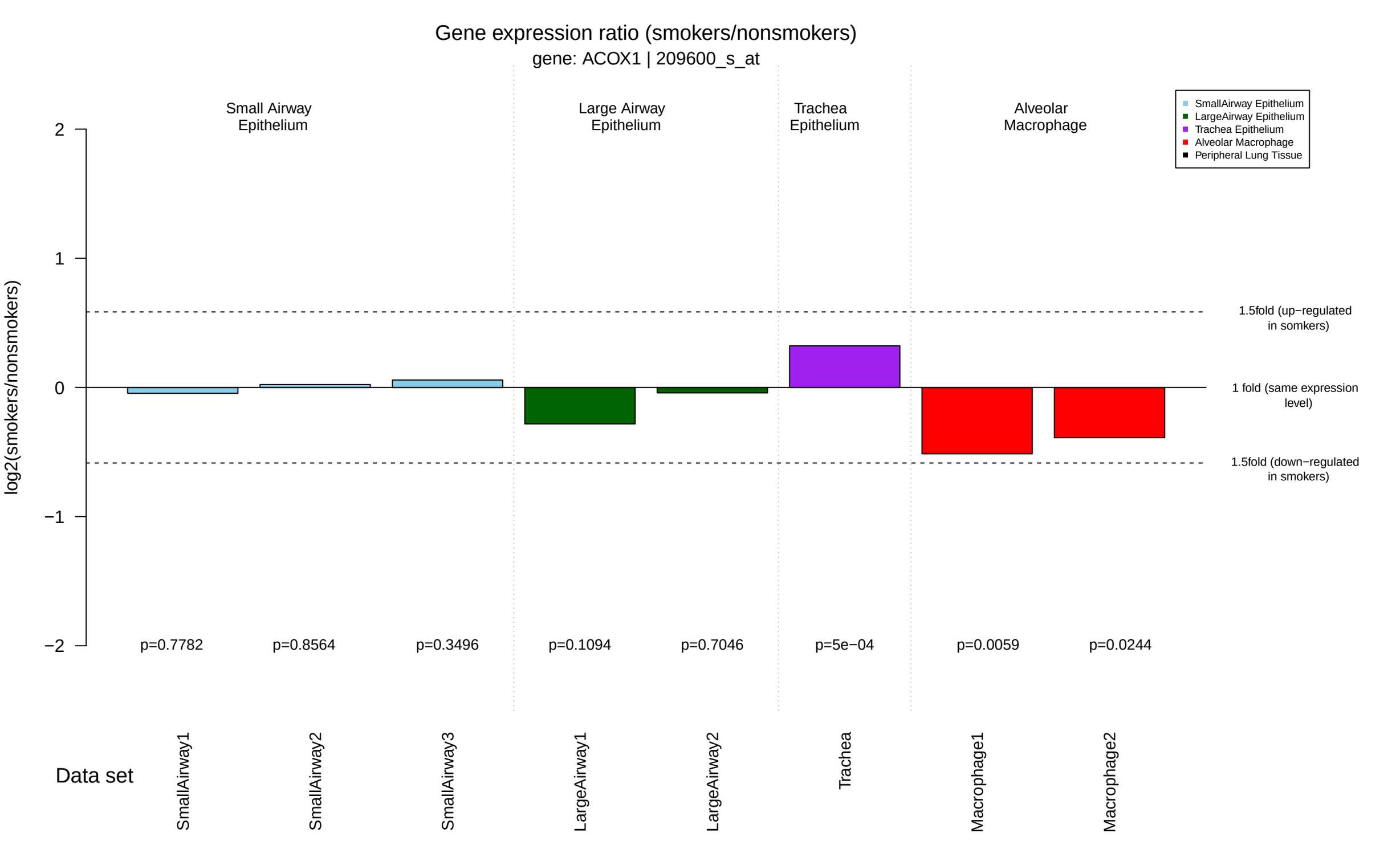
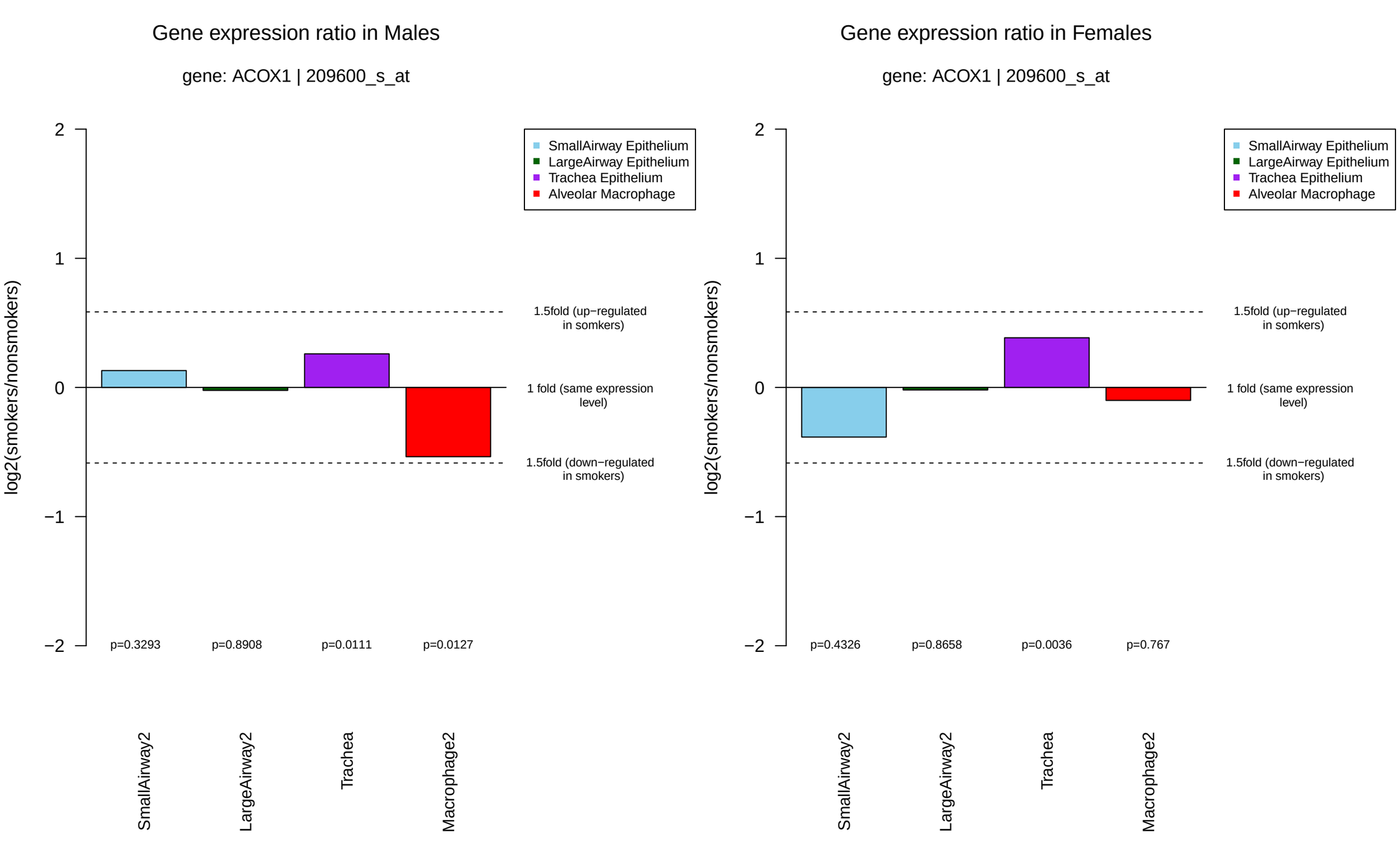
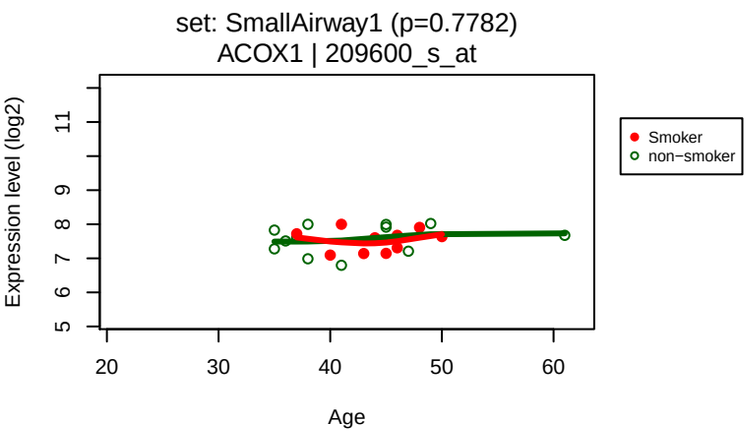
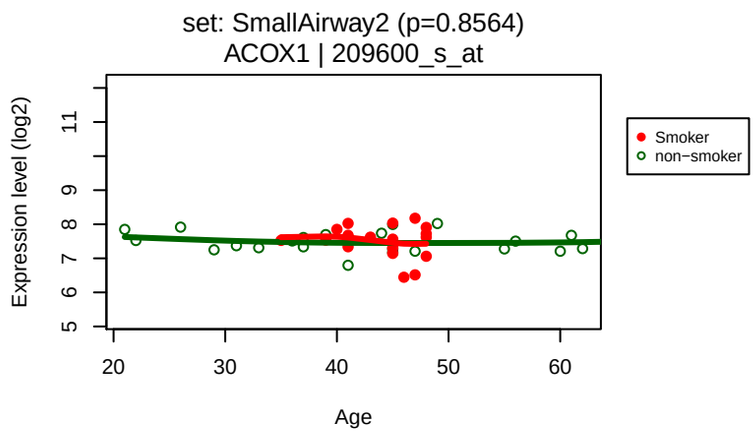
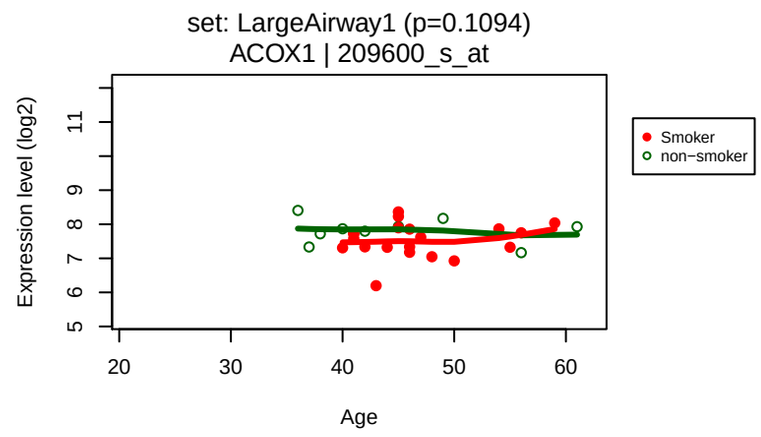

Gene expression of smokers and nonsmokers
gene: ACOX1 | 209600_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7782
p=0.8564
p=0.3496
p=0.1094
p=0.7046
p=5e−04
p=0.0059
p=0.0244
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

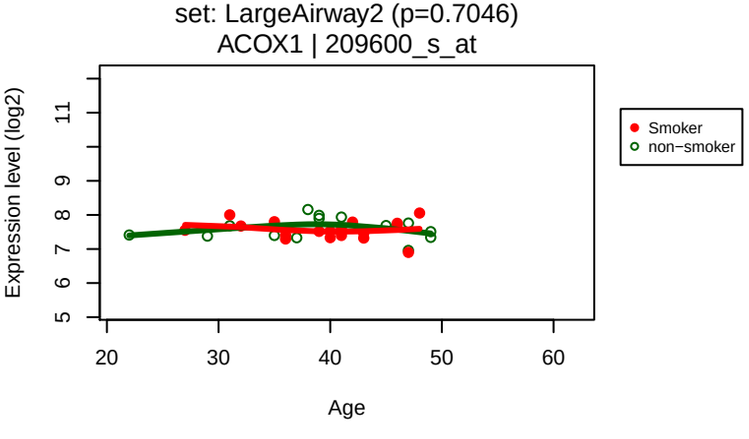

Gene expression ratio (smokers/nonsmokers)
gene: ACOX1 | 209600_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7782
p=0.8564
p=0.3496
p=0.1094
p=0.7046
p=5e−04
p=0.0059
p=0.0244
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

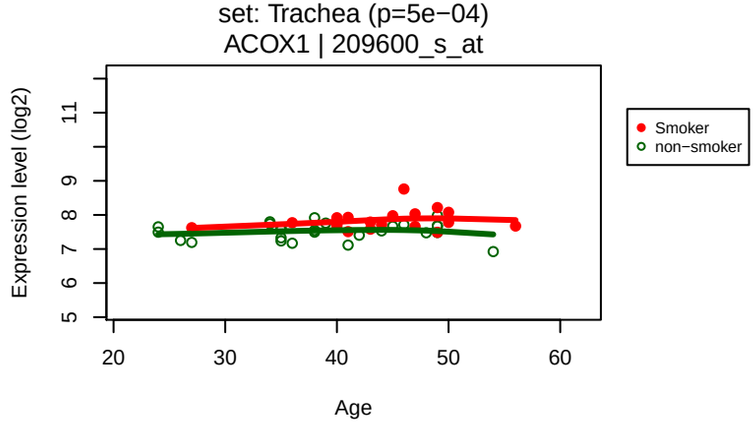

Gene expression ratio in Males
gene: ACOX1 | 209600_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3293
p=0.8908
p=0.0111
p=0.0127
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACOX1 | 209600_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4326
p=0.8658
p=0.0036
p=0.767
SmallAirway2
LargeAirway2
Trachea
Macrophage2

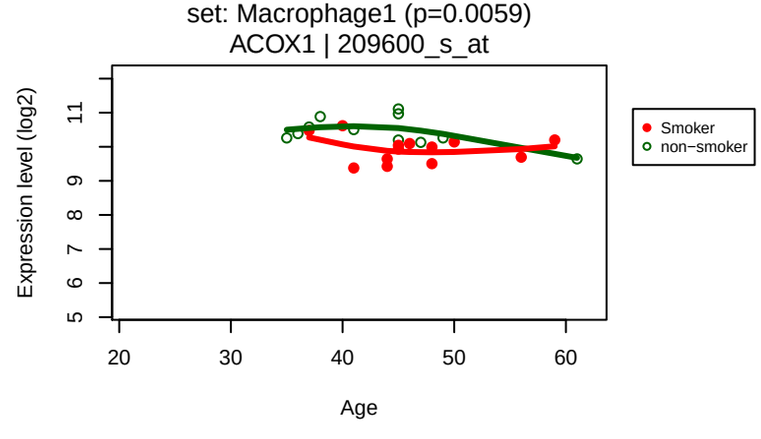

set: SmallAirway1 (p=0.7782)
ACOX1 | 209600_s_at
Expression level (log2)
Age
Smoker
non−smoker

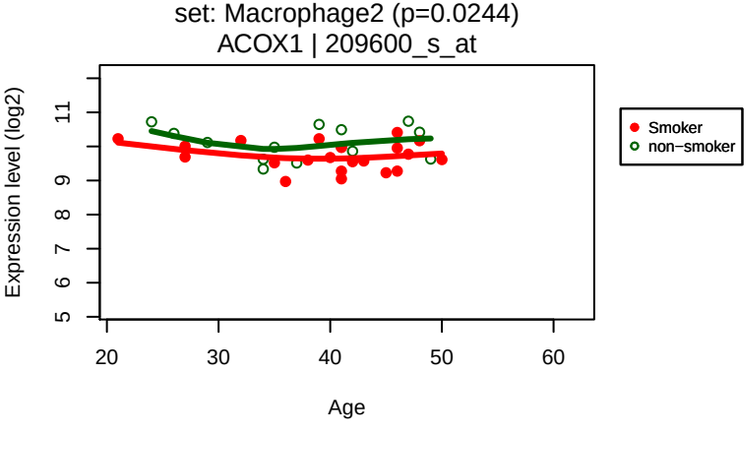

set: SmallAirway2 (p=0.8564)
ACOX1 | 209600_s_at
Expression level (log2)
Age
Smoker
non−smoker

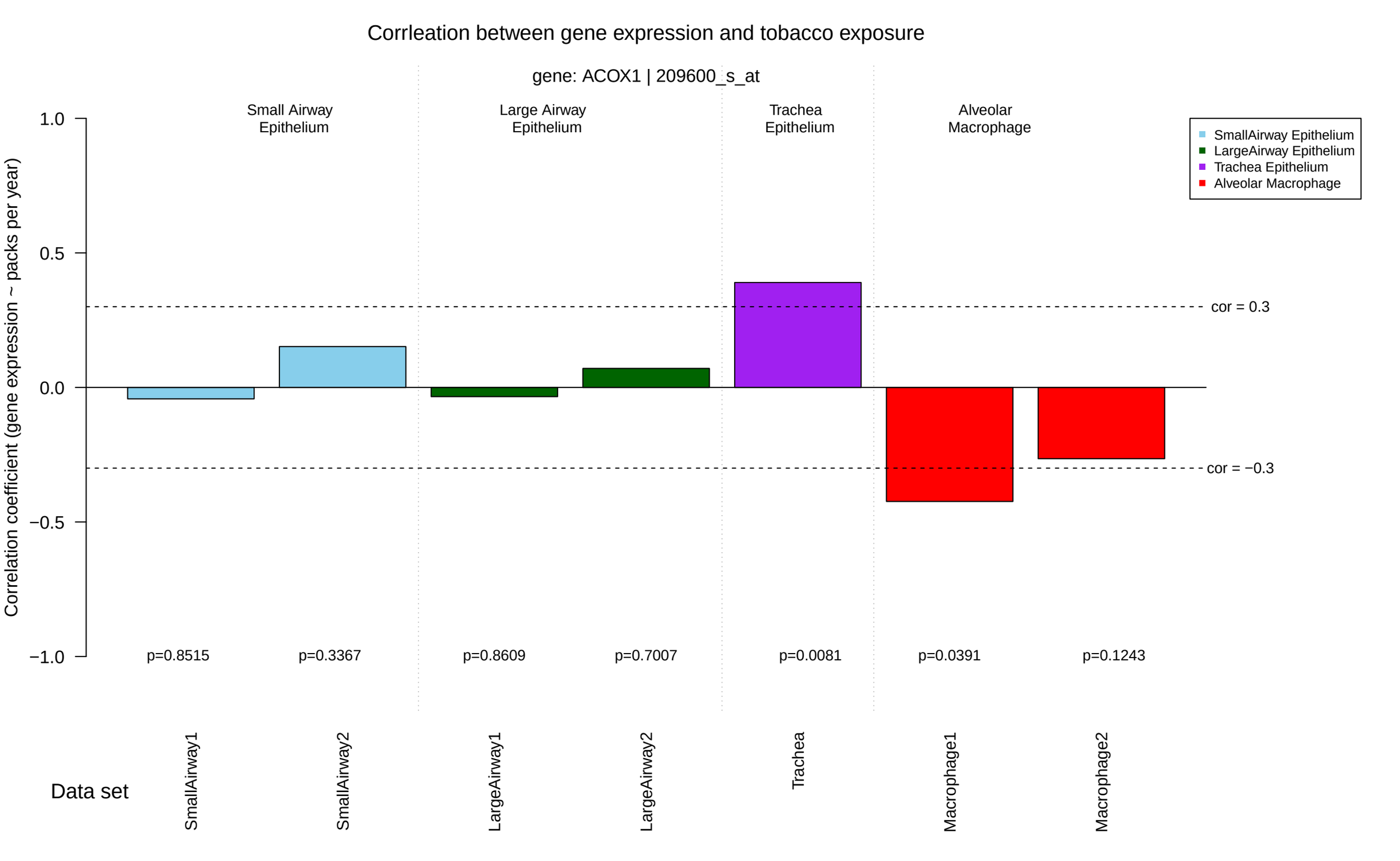

set: LargeAirway1 (p=0.1094)
ACOX1 | 209600_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7046)
ACOX1 | 209600_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
ACOX1 | 209600_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0059)
ACOX1 | 209600_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0244)
ACOX1 | 209600_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACOX1 | 209600_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8515
p=0.3367
p=0.8609
p=0.7007
p=0.0081
p=0.0391
p=0.1243
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2